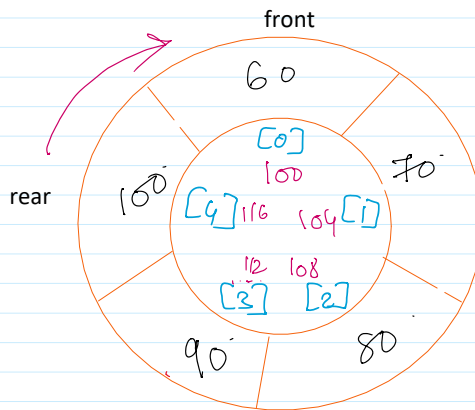


int arr[5]

=-1

Queue Full Condition



$$\begin{aligned} \text{front} &= (\text{rear} + 1) \% \text{SIZE} \\ 1 &= (0 + 1) \% 5 = 1 \\ 2 &= (1 + 1) \% 5 = 2 \\ 3 &= (2 + 1) \% 5 = 3 \\ 4 &= (3 + 1) \% 5 = 4 \\ \underline{\underline{0}} &= (4 + 1) \% 5 = \underline{\underline{0}} \end{aligned}$$

Enqueue:

- 1) Increment rear as
Rear = (rear+1) + SIZE
1. Add element at rear position
2. If front == -1, make front = 0

Dequeue:

- 1) Increment front .
- *****

If front is 4 and rear is 0
to delete the rear position, we cannot
increment front as front ++ Will be
index 5 but we want to delete index 0;
So,
Front = front + 1 % SIZE

If deleting the last element in queue
If(front == rear)
Front = rear = -1

Dequeue()

$$\begin{aligned} \text{① arr}[\text{front}] &= 0; \\ \text{front} &= (\text{front} + 1) \% \text{SIZE}; \end{aligned}$$

Queue Empty condition

If(rear == -1) queue is empty

Queue Full condition :

When Queue is full,

Rear = 0 front 1
Rear = 1, front = 2
Rear = 2 front = 3
Rear = 3 front = 4
rear = 4, front = 0

This means,
Front == rear + 1

But when rear = 4 front = 0
0 == 4+1 does not satisfy the above condition
So the queue full condition can be
Front == (rear+1) % SIZE
0 == (4+1) % 5
0 == 5%5
0==0

Enqueue.

- ① rear++;

$$\text{rear} = (\text{rear} + 1) \% \text{SIZE};$$

$$\begin{aligned} (-1+1) \% 5 &= 0 \\ \text{rear} &= (0+1) \% 5 = 1 \\ \text{rear} &= (1+1) \% 5 = 2 \\ \text{rear} &= (2+1) \% 5 = 3 \\ \text{rear} &= (3+1) \% 5 = 4 \\ \underline{\underline{\text{rear}}} &= (4+1) \% 5 = \underline{\underline{0}} \end{aligned}$$

Option 2:

If rear == SIZE-1
Rear = 0;
Else
Rear++;